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CHALLENGES WITH TERTIARY-LEVEL MECHATRONIC FLUID POWER

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ABSTRACT

We take the view that mechatronics, as it relates to fluid power, has three levels which we designate as primary, secondary and tertiary. We briefly review the current status of fluid power, hydraulic and pneumatic, and of electronic control of it. We then focus on tertiary-level mechatronic fluid power and the challenges to it being applied successfully.

KEYWORDS

Fluid Power, Mechatronics, Hydraulic Control, Pneumatic Control

INTRODUCTION

We briefly present our views of the current status of fluid power itself, of mechatronics, system design, control, and artificial intelligence, as they relate to fluid power systems. This provides the background we feel to be necessary to expose the practical challenges to be met before practical tertiary-level mechatronic fluid power systems can be realised. We attempt to expose the current difficulties in meeting these challenges.

ABOUT FLUID POWER

Use of pressurised fluids to provide powered actuation, ie. integrated force and motion, goes back a long time. Use of compressed air to power automation devices remains current. There are hand tools such as jackhammers and drills, air powered brakes, simple raising and lowering, pushing or pulling, clamping and unclamping operations of moderate power very amenable to economic and effective pneumatic power for their actuations, using compressed air at around 0.7 MPa (100 psi). With the advent of automatic control in industry came an array of low-pressure (0.1 MPa, 15 psi) pneumatic sensors, controllers, logic and small-power actuation devices. These were very widely used in industrial automation, their only real rival being electrical relay-type counterparts. Electronics, as it developed reliability and low cost, pushed both low-pressure pneumatic and electrical relay controller technologies into the background, without inhibiting the use of pneumatics as a power technology. In fact the marriage of electronic control and pneumatic power is very successful and enduring in industrial automation. Pneumatics has motors and cylinders as actuators. It retains a modest industrial logic capability, an attraction being the direct integration of this modest logic capability with moderate actuation power. Integration of PLC's with pneumatic power is especially successful. A number of major multinational companies provide modern pneumatic and electro-pneumatic devices and systems for industrial automation.

Hydraulics is the second and more powerful arm of the fluid power duo. It has easily recognised roots in the centrally-pressurised water reticulation systems supplying power to industrial machines of the 19th Century. From about 100 years ago, hydraulics using oil in stand-alone systems began to replace the reticulated water powered systems, for modern mobile equipment, aircraft and large machines. Working pressures increased over the years, and now 20 MPa (3000 psi) is general and 30 MPa (4,500 psi) common. The last three decades have seen increased use of artificial and water based fluids in situations of fire risk or product contamination. There is also new hydraulics technology which allows the use of tap water as the hydraulic fluid medium, a full circle situation. Hydraulics, like pneumatics, allows the use of motors for continuous rotary motion, cyliners for rectilinear motion, and rotary actuators for limited rotary motion. Also like pneumatics, hydraulic actuation has increasingly integrated with electrical and then electronic control.

The circuits of fluid power systems begin with the source of power (compressed air supply; hydraulic pump) and end with powered actuators (motors, cylinders, rotary actuators). In between are valves for

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the control of pressure, flow-rate, and flow direction, and equipment for conditioning the fluid (filters, etc) and monitoring system state. There can be overall system control ranging from direct manual operation and supervision through electrical remote control of some devices, to on-line computer control and supervision of whole systems.

Both hydraulics and pneumatics are, in 1996, predominantly actuation (muscle) technologies. Though there remains many applications where they are controlled manually, they are increasingly under electronic control. They rival each other only over a narrow power band, at the top end of pneumatic power and the bottom end of hydraulic power. In this grey area of power, they are rivalled also by dc electric actuation. At higher powers, hydraulics can expect to be increasingly rivalled by the developing AC electrical power automation technology, in which sophisticated electronic control brings quality speed control to simple, rugged, inexpensive AC motors.

ABOUT MECHATRONICS

Many definitions of mechatronics exist. They range from the simple "integration of electronics and mechanicals" to tortuous attempts to embrace all possibilities. Figure 1 illustrates our interpretation of the essentials of mechatronics. The global necessities Design, Control and Actuation are generic - they do not imply any particular technique or technology. Within each of these three strategic segments are listed some of the tactical requirements. Figure 1 does not specifically mention things like electronics, computers, computer-aided design, as these things are ubiquitous in 1996. It highlights system conception and the preparation for its practical realisation (ie. Design); the need for control actuation to actually perform the task(s); and the integration of these to realise the useful mechatronic system. All of the actuation technologies involve "mechanicals", even though the primary muscle power might be pneumatic, electric, or hydraulic. Fluid power is clearly a major actuation or muscle technology option in the field of industrial mechatronics.

It should not be thought that all mechatronic systems will contain all or even most of the features listed in or implied by Figure 1. There are simple, medium and sophisticated mechatronic applications, which we refer to as Primary, Secondary, and Tertiary levels. An electronic/electrically-controlled pneumatic or hydraulic valve is a primary level mechatronic system. It is a fairly basic component destined to be a small part of an overall system. Controlled electrohydraulic and electro-pneumatic servosystems are secondary level mechatronic systems, containing some sophisticated control and interfacing between electronics and mechanicals. They may have limited programmability, but no real intelligence is likely. They may be under further supervisor control, either manual or automatic. A tertiary level mechatronic system will have on-line intelligence, thus being more autonomous than primary or secondary level systems. A sophisticated continuous-path industrial robot with on-line state sensing and control is a tertiary-level mechatronic system.

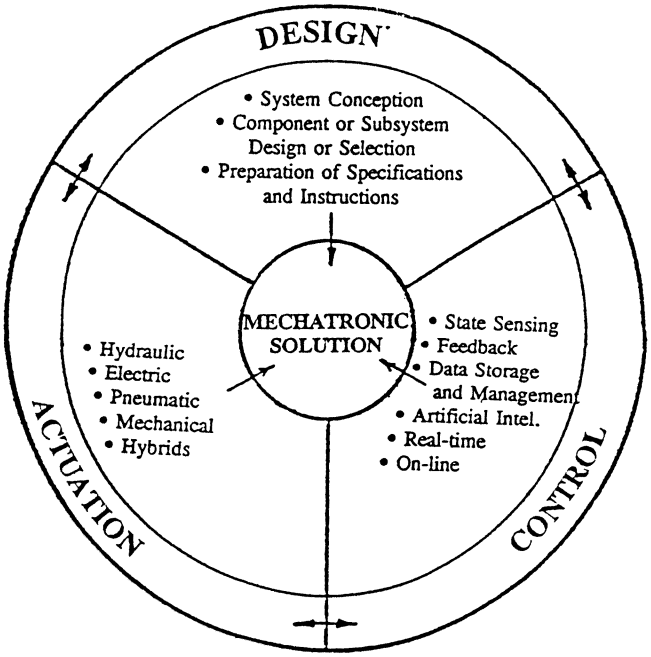


FIG. 1 INGREDIENTS FOR TERTIARY MECHATRONICS

ABOUT SYSTEM DESIGN

Design is the creative aspect of engineering and the conception and specification of realistic, reliable, economic, and well performing mechatronic fluid power systems fits under this umbrella. Most here will realise that the design of fluid power systems is a matter of judicious selection of off-the-shelf fluid power components, and their assembly into an appropriate working system matched to the tasks to be performed. Simple fluid power systems for simple tasks can be "designed" by personnel with appropriate seat-of-the-pants experience. Where the task is simple, the load not very large, and the dynamics of the motions not very significant, such designs will usually be effective.

Where closely controlled actuation of significant resisting loads is required, the dynamics are important and usually dominant in the quality of a system's performance. It is not enough to know that a plant or

load's state can be changed from state A to state B by the control system. The quality of state during the change is always important and can be vital. How long does it take to change state? How does state fluctuate during the change? Are the plant and its control system, acting together, stable? Ensuring good dynamic response is surely a necessary part of design. This is particularly the case for heavily loaded hydraulically actuated systems, where the quality of response is largely dictated by the R (resistive), C (capacitive) and I (inertive) effects associated with the driven load. The state sensing, decision making, and action instructions can be effected very quickly by modern electronics and computer controls, but the controlled plant's R,C and I effects will dictate how long any change-of-state will take and largely its quality, though the latter can be corrected if necessary by on-line control techniques.

The quality of response of the overall system is best ensured by dynamic modelling and simulation of the system during its design. The more complex the system, the more such simulation is desirable if expensive mistakes are to be avoided or at least minimised. There are commercial computer packages available to assist. Some are general simulation packages, where the designer-analyst separately develops a proposed system's model and the model's coefficient values. ACSL and CSMP are well known. Their newest versions allow windows-type interaction for changing components models or coefficient values. Others are technology orientated; Bathfp, HOPSAN and DSHplus are for fluid power control and drive systems. They allow interaction with the designer-analyst in a windows mode while systems are being designed. They provide options for dynamic models of common components used in fluid power systems. There are new user-friendly packages such as VISSIM which feature engineering generality rather than being technology specific.

Such simulation as a design aid can be used to help optimise primary level mechatronic devices, and to ensure satisfactory performance of whole secondary or tertiary level systems. A complete system should have good individual components. However, good components of itself does not guarantee a good system. The dynamic interactions between the components or sub-systems will dictate the quality of the overall system performance.

ABOUT CONTROL

For years the ideas and theories for improving the quality of automatic control outstripped the capacity to apply them in on-line real-time situations. Research literature is heavy with potential applications for adaptive control, optimal control and intelligent control. There have been limited applications of aspects of these in high-cost space, aircraft, and military areas, with some spin-off into industrial machines such as industrial robots and machining centres. It is fairly recent that computer power increase and cost reduction allied with similar developments in on-line state monitoring have encouraged direct use of sophisticated control techniques on industrial machinery in general. Tertiary level and some secondary level mechatronic fluid power systems are in this category. The Proceedings of Fluid Power Conferences of the past five years have many examples of prototype mechatronic fluid power systems involving sophisticated on-line control.

ABOUT ARTIFICIAL INTELLIGENCE

Those aspects of Artificial Intelligence (A1) that seems to be relevant and practical for mechatronic fluid power (MFP) are:

- Expert Systems, which hold in tabulated and accessible form the experience of one or more experts in the field. The technique is deterministic in that if certain system state values or trends exist, specific corrective actions are indicated, with a high probability of correct decision making.

- Neural Networks which attempt to reproduce the memory, logic and decision making of the aware human eye-brain combination.

- Fuzzy Logic, which attempts to manage system state, environmental, and command inputs much as the skilled human controller does. Fuzzy logic varies from its parent Boolean logic in that Boolean logic allows a parameter to have only one of two values (ie. true or false), while Fuzzy Logic allows a parameter to have one of a number of values from a set.

Neural networks and fuzzy logic tend to be heuristic rather than deterministic, and thus represent higher level intelligence. Experience, intuition and judgement in the face of undescribed and possibly unpredicted plant behaviour need to be coped with.Genuine AI requires powerful computing systems. On line AI probably requires parallel programming (transputers).

- Generic Algorithms

CHALLENGES

The paper so far has provided background to expose the challenges or difficulties to be met for effective realistic tertiary level mechatronic systems for year

2000 industrial control of operating plant or machines involving fluid power.

We summarise these as:

- understanding the control and load dynamics prior to or as part of system design
- incorporation of state monitoring equipment in the design capable of providing reliable continuous, or discreet at appropriate time interval, data on system operating state
- electronics capable of accepting system state data and generating from it values of state errors and their trends
- electronics capable of making simultaneous multi-variable error-correcting decisions either continuously or at appropriate frequencies
- Appropriately trained people
- Effective realisation of artificial intelligence

With regard to the control of a proposed system's dynamics, there seems to be widespsread inability and/or unwillingness to identify, appreciate, describe and interlink the interacting R,C, and I effects present naturally and by design. The development of adequate dynamic models is rarely easy. It requires insight, training and experience. We strongly recommend use of a structure-based modelling procedure such as block diagrams, signal flow graphs, power port diagrams, bond graphs, rather than attempting to write an equation set model without such structure being first perfected. For powered systems, we strongly recommend the use of power flow modelling procedures because of their fundamental and recognisable association with the real systems structure and characteristics. In particular we recommend bondgraphs for their extension of insight into the system's dynamic nature, and for their ability to cross technology boundaries. Even though the designer-analyst may wish to use a block diagram orientated simulation procedure such as VISSIM, it is safer to develop the model to be simulated via power flow procedures. It is simple enough to convert a bond graph or power port diagram to an equivalent block diagram. In that way, you separate structure development from equation writing, an impossibility with direct block diagram formation.

With regard to state monitoring, and the processing from it of complex error-correcting decisions, it seems that much of the hardware (transducers, interfaces, computers) is available. Also that it is getting low enough in cost to encourage dedicated use if cost-benefit study shows that sophisticated mechatronics is viable. Transputers and the ability to program them efficiently and effectively to provide parallel procesing is an area needing attention.

With regard to the attractive idea of intelligent control using Artificial Intelligence, it seems that

- Expert Systems will provide a useable component of relatively low intelligence and autonomy and must be used with neural nets and/or fuzzy logic to be of use in on-line control. Relative to tertiary level MFP there is need for diagnostic and interpretive expert systems which pool the expertise of a number of accepted experts and which have been tried and culled so that they earn general acceptance by fluid power system designers.
- that Neural Nets, a very attractive concept, are unlikely to be practical in hardware for some time yet. More work also is needed on how to rationally design a neural net component for a particular application, and on how to train neural nets efficiently and quickly.
- that elementary fuzzy logic control is practical now, but that clearer procedures for designing fuzzy logic for particular applications is needed. Arbitrary and trial-and-error procedures seem to be the present way. How do we establish appropriate entries in fuzzy sets, and what rules to follow?

With regard to people, the need is for some truly cross-discipline expertise covering design, electrical-electronic engineering, mechanical engineering, computer science and artificial intelligence. Hands-on experience and expertise is essential. Parallel programming requires expertise in special low-level languages, not quickly mastered. Some of these people need to have the qualifications, authority, drive and willpower to conceive and realize tertiary-level mechatronic fluid power systems.

FURTHER COMMENTS

A neural-net attempts to perform some of the intelligence of the human eye-hand-brain in sensing, recognition, learning, memorising, recalling, weighting of simultaneous inputs, perceptive judgement, and decision making. Neural networking is a computer technique using a large number of inter-connected processing elements which operate in parallel. It is distinctive from conventional computing which relies on a smaller number of more complex processing elements which operate in sequence. At present, most neural-nets are computer package simulations.

Ultimately they will be electronic hardware parallel processors able to effect the massive computational requirements to fully realise the potential of the neural-net philosophy.

There are many forms of neural-nets in the literature, only some of which are relevant to plant or process control. A neural-net, like a human operator, must be trained or taught to operate effectively in particular situations; ie. how to respond acceptably if unpredicted operating conditions arise.

The main challenge in the development of neural net controllers is to economically produce powerful and reliable computers capable of on-line parallel processing. Another challenge is to develop algorithms to induce rapid and reliable learning by neural-nets. Another is to develop rational quantitative ways for designing neural-nets for specific applications; an excessive number of neurons causes slow convergence; insufficient results in inaccurate or false conclusions.

Control of commercial fluid power systems via neural net techniques is not likely for some time. However, use of neural-nets in the monitoring of such systems may become practicable. Research has focused on Expert Systems as the most appropriate AI technique for monitoring the state of working fluid power systems. Expert system technology is not well suited to accepting and interpreting unprocessed sensor signals. Perhaps a neural-net could provide this function for the expert system.

Conventional automatic control systems have controllers with fixed structure and coefficients, and it is usual to assume, often with a blind eye, that the plant being controlled also has fixed parameters. Such systems have no autonomy and therefore no intelligence. They are "trained" to act in well defined deterministic ways. Adaptive controllers allow controller parameters and perhaps structure to change to better accommodate changing plant characteristics and some uncertainty about the time and magnitude of disturbances to the operating system. Usually the range of plant operating variation is known and the controller adaptation is limited to accommodate a situation of fairly high certainty. Even so, a degree of autonomy is present, and the adaptive controller can be thought of as having some intelligence. Truly intelligent control must emulate the ability of the trained human operator in terms of perception, learning, memory, experienced intuition and common sense, judgement and decision making, and adaptation to present circumstances in an environment of partial or complete uncertainty. The intelligent controller, be it human or machine, must be able to recognise and interpret simultaneous multiple data which may be incomplete, inaccurate or noisy, extraneous, even partially wrong or missing, and from it make correcting control decision with a very high probability of being correct.

CONCLUSION

We have attempted to provide a background, a present, and a foreground as a contribution to the ongoing debate about where sophisticated on-line automatic control of fluid power actuation might be heading.